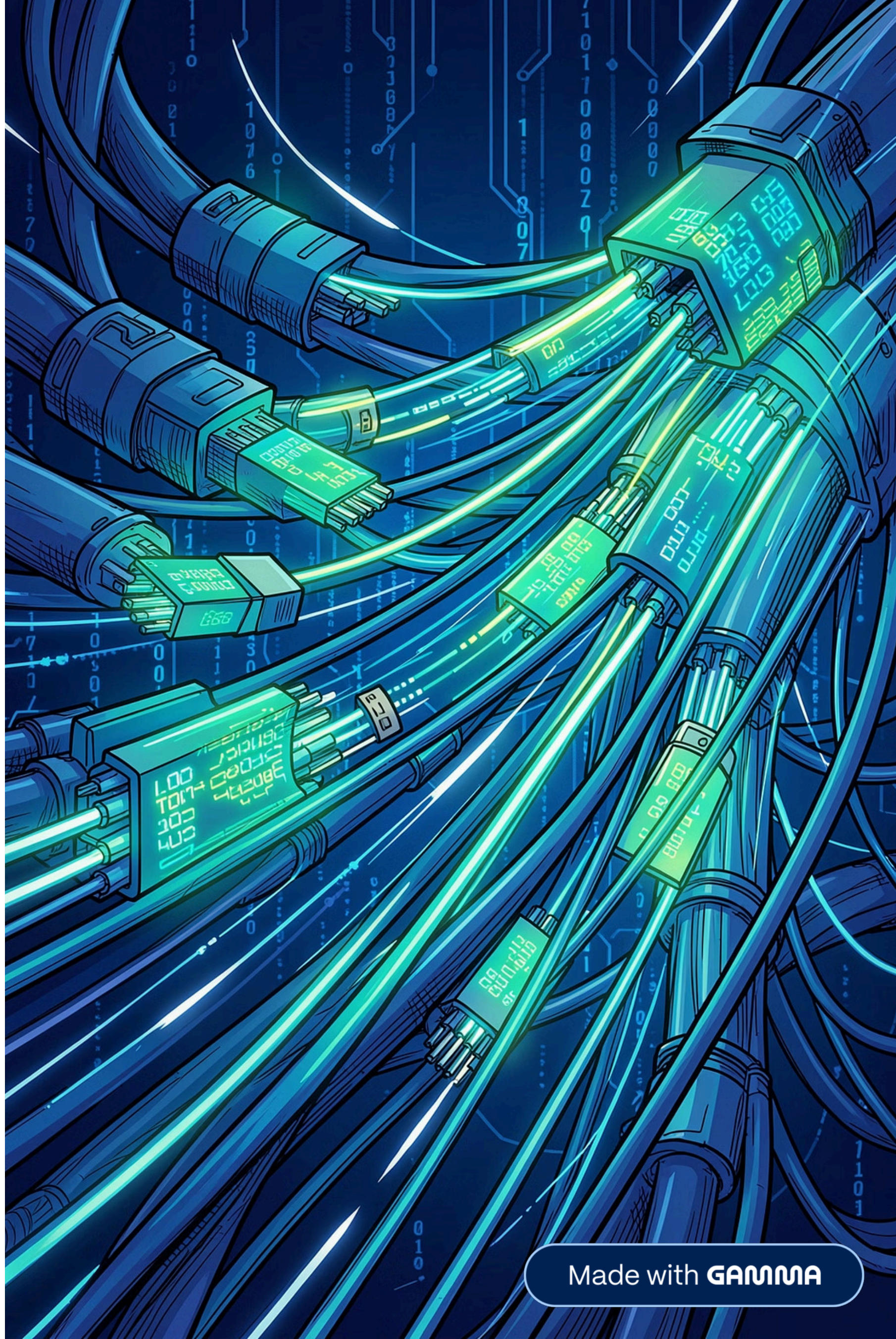
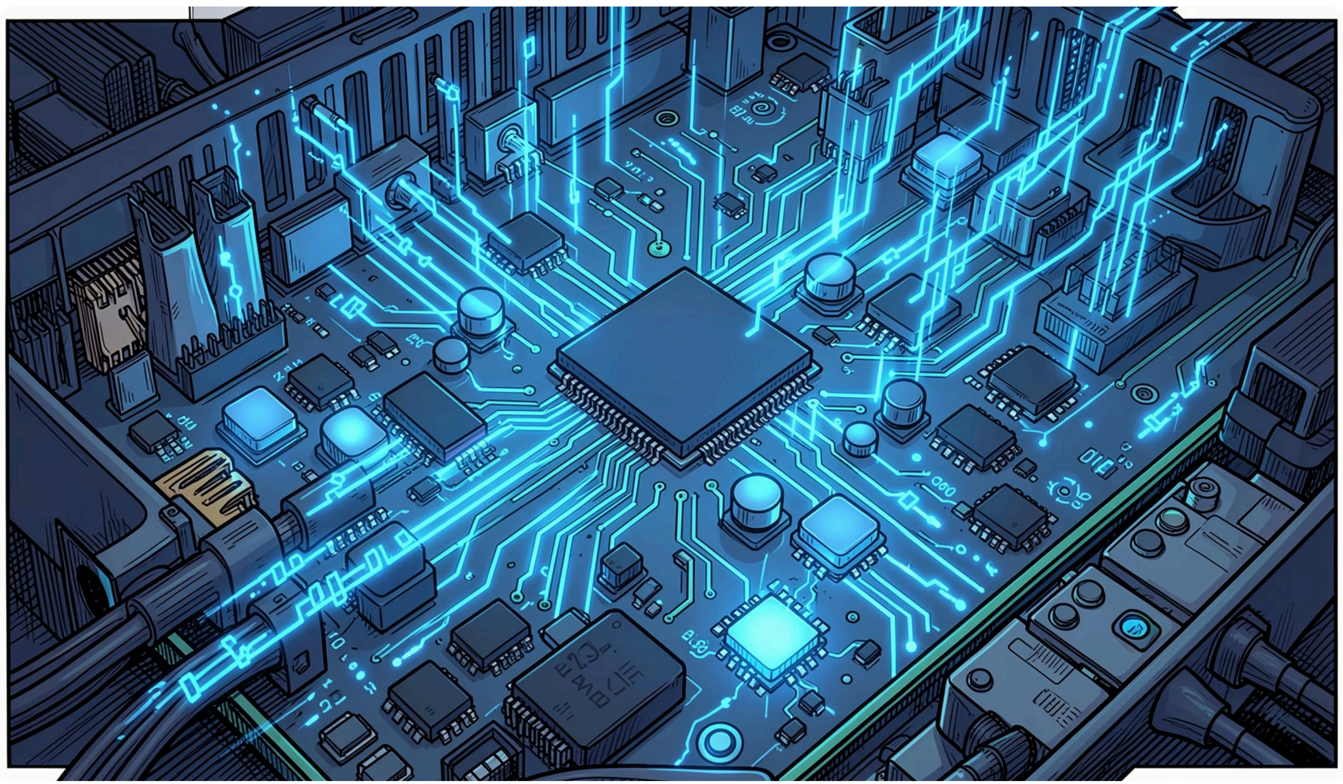
IPv4 Header Fields

A comprehensive breakdown of every field in the IPv4 packet header — what each field does, how it works, and why it matters for network communication.



FIELDS 1 & 2

Version & Internet Header Length



Version (4 bits)

Indicates the IP version being used. For IPv4, the value is always **4**.

Purpose: Helps devices identify whether the packet is IPv4 or IPv6.

Internet Header Length — IHL (4 bits)

Specifies the length of the IPv4 header, measured in **32-bit words**. Minimum value = 5 (20 bytes).

Purpose: Tells the receiver where the actual data begins in the packet.

FIELDS 3 & 4

Type of Service & Total Length

Type of Service / DSCP

Used to assign packet priority. Critical for **Quality of Service (QoS)**.

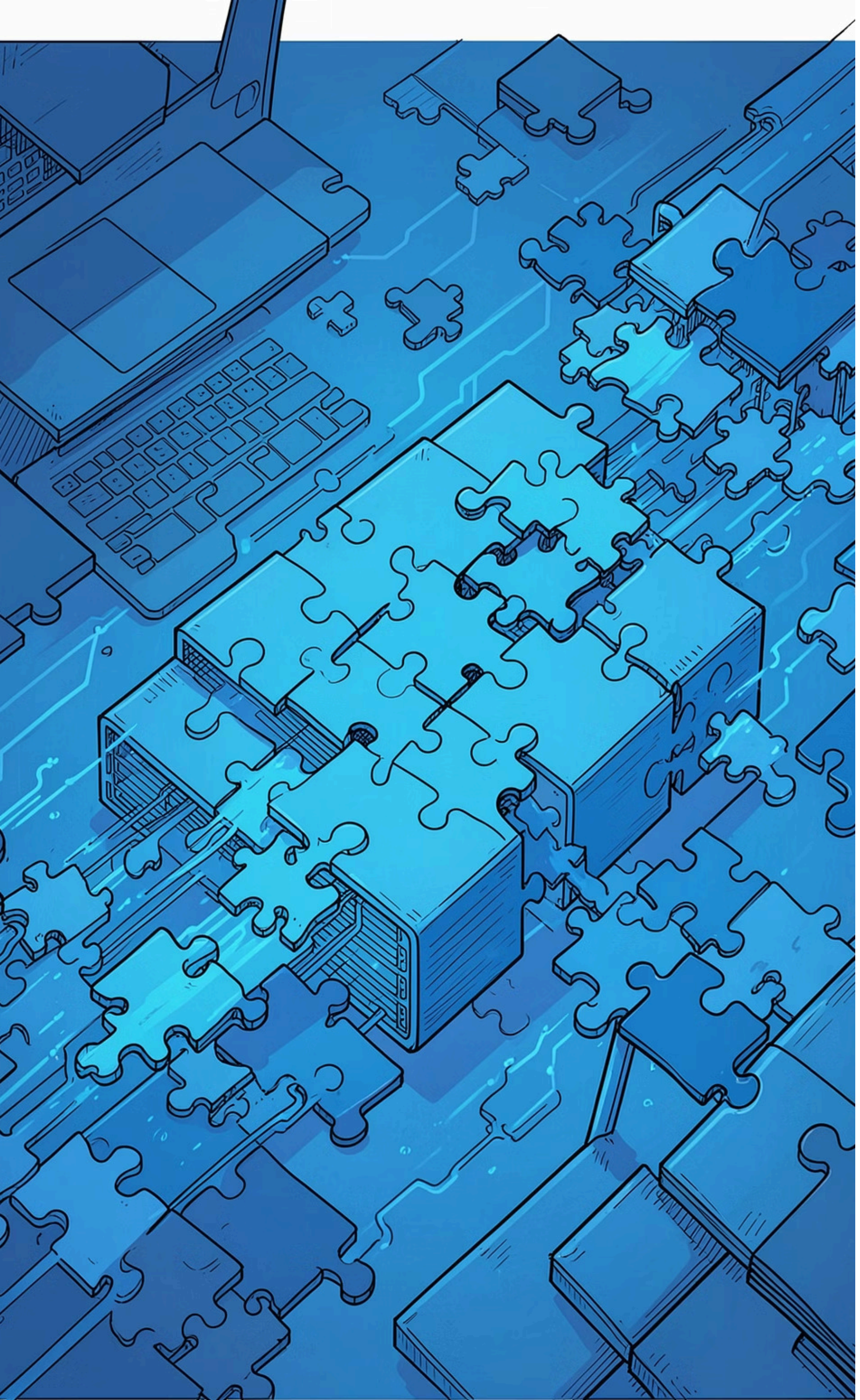
These packets can be prioritized over normal traffic:

- Voice calls
- Video conferencing
- Streaming

Total Length (16 bits)

Specifies the total size of the packet, including both the **header** and the **data**.

Maximum Size: 65,535 bytes



FIELDS 5, 6 & 7

Identification, Flags & Fragment Offset

Identification (16 bits)

A unique number assigned to each packet. When a packet is fragmented, all fragments carry the same Identification number so they can be reassembled correctly at the destination.

Flags (3 bits)

Controls fragmentation behavior.

DF (Don't Fragment): DF=1 → cannot fragment; DF=0 → allowed

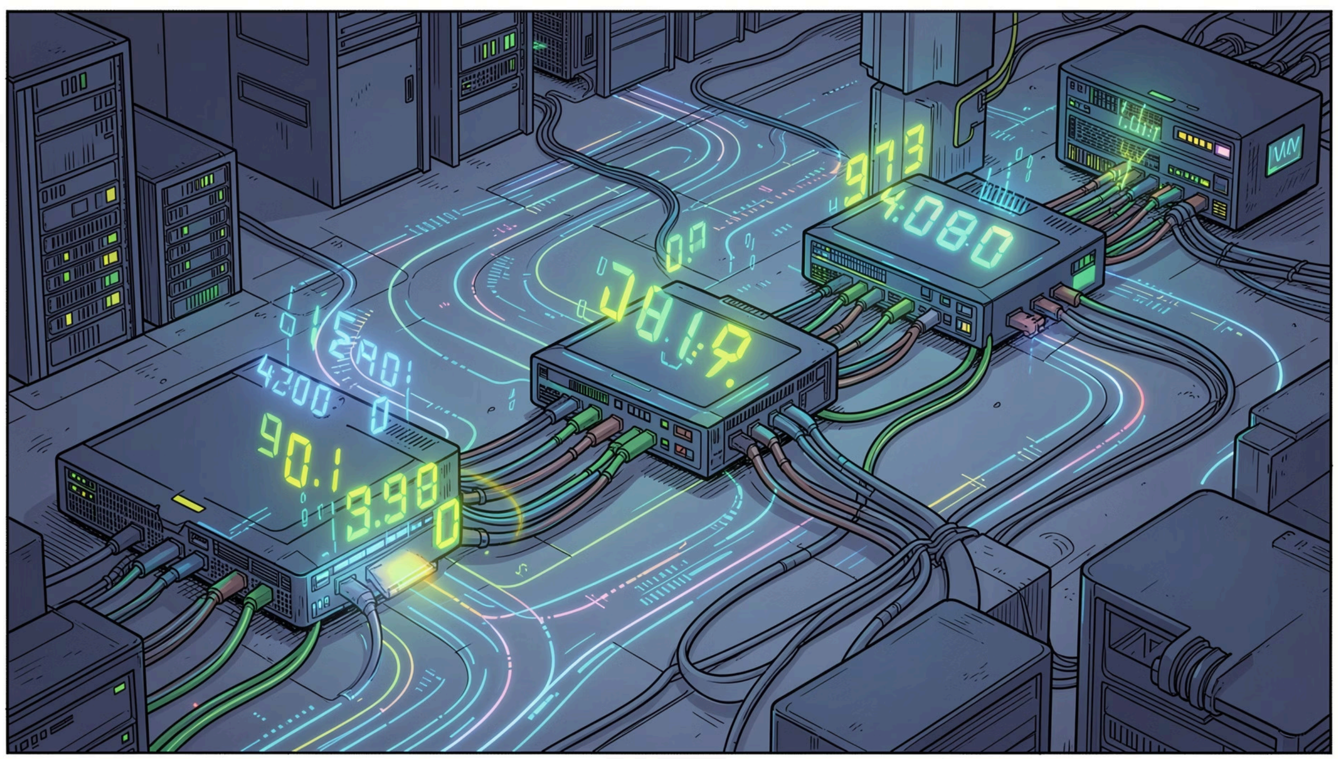
MF (More Fragments): MF=1 → more coming; MF=0 → last fragment

Fragment Offset (13 bits)

Indicates the position of a fragment within the original packet. Helps the receiver reassemble fragments in the correct order.

FIELDS 8 & 9

Time To Live & Protocol



Time To Live — TTL (8 bits)

Determines how long a packet can stay in the network. Decreased by 1 at every router.

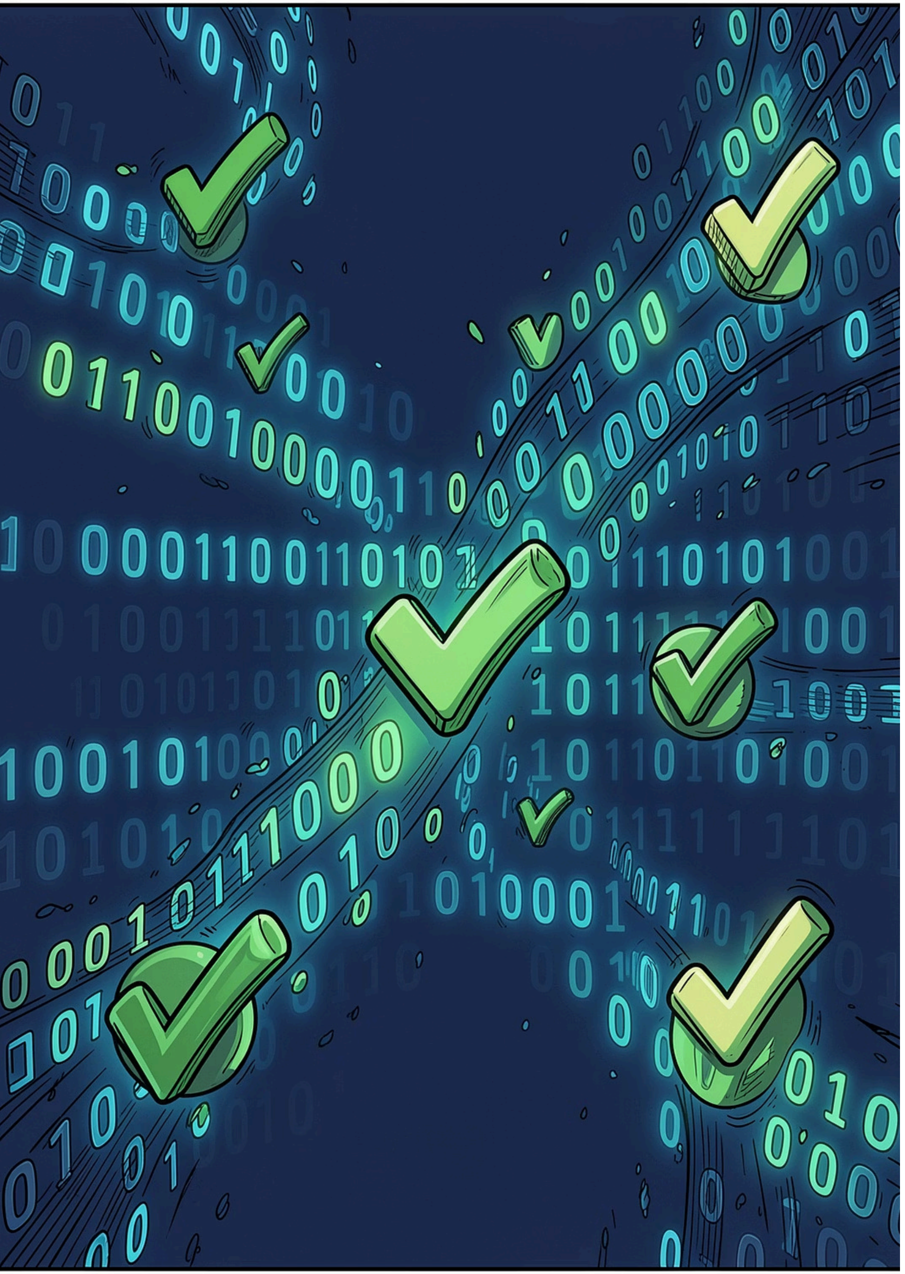
Example: Initial TTL = 64 → Router 1 = 63 → Router 2 = 62 → Router 3 = 61. When TTL reaches 0, the packet is discarded.

Purpose: Prevents infinite routing loops.

Protocol (8 bits)

Identifies which upper-layer protocol receives the data. Allows IP to deliver data to the correct transport protocol.

Value	Protocol
1	ICMP
6	TCP
17	UDP



FIELD 10

Header Checksum

Header Checksum (16 bits)

Detects errors in the IPv4 header. Recalculated at every router.

Purpose: Ensures header integrity during transmission.

FIELDS 11 & 12

Source & Destination Address

Source Address (32 bits)

Contains the **sender's** IPv4 address.

Example: 192.168.1.10

Destination Address (32 bits)

Contains the **receiver's** IPv4 address.

Example: 8.8.8.8

FIELDS 13, 14 & DATA

Options, Padding & Payload

Options (Variable Length)

Provides additional features such as security, route recording, and timestamping. Usually not used in normal communication.

Padding

Extra bits added to make the header length a multiple of 32 bits. **Purpose:** Maintains proper alignment.

Data (Payload)

Contains the actual information being transmitted. Examples: web page data, email content, video stream, file transfer.